Introduction 1

This technical report does not attempt to cover the entire scope of polypharmacy, but merely aims to introduce polypharmacy as a concept, and examine some approaches for the appropriate management of polypharmacy, which are crucial for ensuring greater medication safety.

1.1 Polypharmacy

Despite the increasing prevalence of polypharmacy, the term continues to lack a clear universal definition. A recent systematic review of the definitions of polypharmacy showed that the term was most commonly applied to situations where patients took five or more medications, and this numerical definition was used by 46.4% of the studies evaluated *(7)*. Furthermore, there are inconsistencies with the use regarding duration of therapy and whether to include over-the-counter (OTC), and traditional and complementary medicines in the definition or not. However, with the aim of reducing medication-related harm, it is important to verify to the fullest extent possible all the medications that the patient is taking, including all OTC, and traditional and complementary medicines.

Polypharmacy is the concurrent use of multiple medications. Although there is no standard definition, polypharmacy is often defined as the routine use of five or more medications. This includes over-the-counter, prescription and/or traditional and complementary medicines used by a patient (see Annex 1).

While polypharmacy is often defined as routinely taking a minimum of five medicines, it is being more frequently suggested that the emphasis should be on evidenced-based practice (*7*). The goal should be to reduce inappropriate polypharmacy (irrational prescribing of too many medicines) and to ensure appropriate polypharmacy (rational prescribing of multiple medicines based on best available evidence and considering individual patient factors and context) (*7–11*). Therefore, appropriate polypharmacy should be considered at every point of initiation of a new treatment for the patient, and when the patient moves across different health care settings.

Appropriate polypharmacy is present, when (a) all medicines are prescribed for the purpose of achieving specific therapeutic objectives that have been agreed with the patient; (b) therapeutic objectives are actually being achieved or there is a reasonable chance they will be achieved in the future; (c) medication therapy has been optimized to minimize the risk of adverse drug reactions (ADRs); and (d) the patient is motivated and able to take all medicines as intended *(12)*.

Inappropriate polypharmacy is present, when one or more medicines are prescribed that are not or no longer needed, either because: (a) there is no evidence based indication, the indication has expired or the dose is unnecessarily high; (b) one or more medicines fail to achieve the therapeutic objectives they are intended to achieve; (c) one, or the combination of several medicines cause ADRs, or put the patient at a high risk of ADRs or because (d) the patient is not willing or able to take one or more medicines as intended *(12)*.

Polypharmacy has been described as a significant public health challenge *(13)*. It increases the likelihood of adverse effects, with a significant impact on health outcomes and expenditure on health care resources *(14–16)*. Although co-prescribing multiple medicines increases the risk of adverse events, it is important to note that assigning a numeric threshold to define polypharmacy is not always useful. There are cases where polypharmacy is necessary and beneficial, such as the secondary prevention of myocardial infarction, which requires the use of four different classes of medications (a beta blocker, a statin, an antiplatelet agent and an ACE inhibitor) *(8, 12)*. Appropriate polypharmacy recognizes that patients can benefit from multiple medications if the patients' clinical conditions, comorbidities, allergy profiles, the potential drug–drug and drug–disease interactions are considered, and the medicines are prescribed based on the best available evidence *(17)*. Thus, it is critical to distinguish appropriate polypharmacy from inappropriate polypharmacy *(12)*.

The most vulnerable patient groups to the risks of polypharmacy are susceptible to events such as drug–drug interactions, higher risk of falls, ADRs, cognitive impairment, non-adherence and poor nutritional status *(18–20)*. Vulnerable patient groups often include older patients above the age of 65 years and patients who are living in care homes *(19–21)*.

1.2 Prevalence of polypharmacy

Polypharmacy is a major and growing public health issue occurring within all health care settings worldwide *(8, 13)*. The issue is well described in literature from countries in North America *(10, 18)*, Europe *(8, 22)* and the Western Pacific *(23)*, with more data becoming available from other countries in recent years. Additional information from selected countries is available in Annex 2, illustrating that polypharmacy is a global problem. However, variation in the structure of health care delivery and data collection systems, compounded by the different operational definitions of polypharmacy, makes country comparison difficult.

Multimorbidity is defined as the presence of two or more long-term health conditions, which can include (a) defined physical and mental health conditions such as diabetes or schizophrenia; (b) ongoing conditions such as learning disability; (c) symptom complexes such as frailty or chronic pain; (d) sensory impairment such as sight or hearing loss; and (e) alcohol and substance misuse (see Annex 1).

While its true magnitude is not known, the prevalence of polypharmacy is expected to rise due to a multitude of factors *(8)*. First, the global population faces a demographic shift with the proportion of older population

groups on the rise. It has been estimated that the global population aged over 65 years will double from 8% in 2010 to 16% in 2050 *(24)*. In 2015, approximately 5% of the population in OECD countries were aged 80 years and above, this percentage is expected to rise more than double by 2050 *(25)*. Second, epidemiological data indicates that multimorbidity increases markedly with age. In a Scottish study, multimorbidity was prevalent in 81.5% of individuals aged 85 years and over, with a mean number of 3.62 morbidities *(26)*. Ornstein et al. found that the most prevalent chronic conditions in primary care were hypertension (33.5%), hyperlipidemia (33.0%), and depression (18.7%) *(27)*. The presence of multiple morbidities is associated with multiple symptoms, impairments and disabilities. Multimorbidity may result in a combined negative effect on physical and mental health, and can have a major impact on a person's quality of life, limiting daily activities and reducing mobility *(28, 29)*. The need to take multiple medications can be just as problematic, resulting in frequent health care contacts and an increase in the likelihood of medication-related harm *(30)*. Furthermore, it imposes a large economic burden due to patients' complexity of health care needs and frequent interaction with health services, which may be fragmented, ineffective and incomplete *(31)*.

Despite extensive advances in pharmacotherapy, the availability of clinical guidelines for older adults with multiple morbidities is limited *(28)*. Prescribing is largely based on evidence-based guidance for single diseases, which does not generally take into account multimorbidity *(21, 28, 32)*. Consequently, patients are often prescribed several medicines recommended by a number of specialists using disease-specific guidelines which in combination makes the management of any single disease challenging and may even lead to patient harm *(21, 26)*.

1.3 Economic impact of polypharmacy

Advancing the responsible use of medicines: applying levers for change identified several opportunities to reduce health care spending through more responsible use of medicines worldwide. The authors estimated that mismanaged polypharmacy contributed to 4% of the world's total avoidable costs due to suboptimal medicine use. A total of US$ 18 billion, 0.3% of the global total health expenditure could be avoided by appropriate polypharmacy management. Some of the specific recommendations outlined in the report included *(33)*:

- investment in medical audits targeting older patients with multiple medications;
- support for a greater role of pharmacists in medication management and in collaboration with health care professionals for review of therapeutic plans;
- identification of high-risk patients and preparation of targeted medicine management plans for this group; and
- to establish a system for blame-free reporting of medication errors.

The objectives of polypharmacy management should be comprehensive, addressing such issues as improved health outcomes for the patient and population, greater patient engagement in therapeutic decision-making and cost-effectiveness of health care systems and resources. This comprehensive approach – embracing care, health and cost – has been termed the "triple aim" strategy, and it is designed to guide health system performance optimization *(34)*. The term has been built upon to become a "quadruple aim" upon recognizing that staff wellbeing is essential to ensure good care for the public *(35)*.

Improved safety and quality leading to economic benefits

In a randomized control trial by Gillespie et al. clinical pharmacists performed comprehensive medication reviews (see Annex 1) of older hospitalized patients. Patients that received